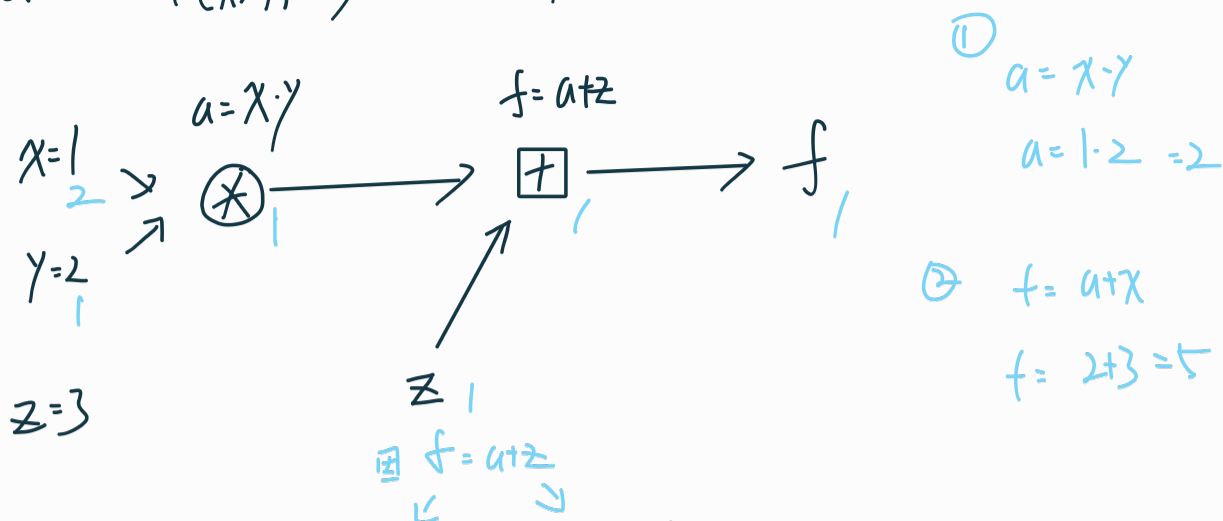


Q₁: $f(x, y, z) = (x \cdot y) + z$, $x=1$ $y=2$ $z=3$



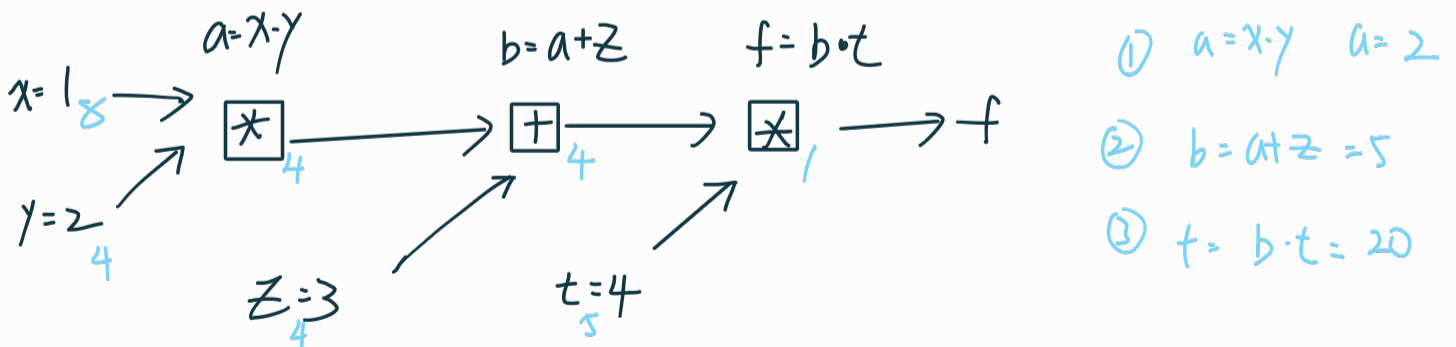
① $\frac{\partial f}{\partial x} = 1$ ② $\frac{\partial f}{\partial a} = 1$ ③ $\frac{\partial f}{\partial z} = 1$

$a = x \cdot y$

④ $\frac{\partial f}{\partial x} = \frac{\partial f}{\partial a} \cdot \frac{\partial a}{\partial x} = 1 \cdot 2 = 2$ $\Rightarrow \frac{\partial a}{\partial x} = y$, $\frac{\partial a}{\partial y} = x$

⑤ $\frac{\partial f}{\partial y} \cdot \frac{\partial f}{\partial a} \cdot \frac{\partial a}{\partial y} = 1 \cdot 1 = 1$

Q₂. $f(x, y, z, t) = ((x \cdot y) + z) \cdot t$, $x=1$, $y=2$, $z=3$ $t=4$



① $\frac{\partial f}{\partial t} = 1$ ② $\frac{\partial f}{\partial b} = t = 4$ ③ $\frac{\partial f}{\partial t} = b = 5$

$\Rightarrow t = 1 \cdot t$ so $\frac{\partial f}{\partial b} = t$, $\frac{\partial f}{\partial t} = b$

④ $\frac{\partial f}{\partial a} = \frac{\partial f}{\partial b} \cdot \frac{\partial b}{\partial a} = 4 \cdot 1 = 4$ ⑤ $\frac{\partial f}{\partial z} = \frac{\partial f}{\partial b} \cdot \frac{\partial b}{\partial z} = 4 \cdot 1 = 4$

$$\textcircled{6} \frac{\partial t}{\partial x} = \frac{\partial t}{\partial a} \cdot \frac{\partial a}{\partial x} = 4 \cdot 2 = 8$$

$$\textcircled{7} \frac{\partial t}{\partial y} = \frac{\partial t}{\partial a} \cdot \frac{\partial a}{\partial y} = 4 \cdot 1 = 4$$

自己對自己用影響變化率一定是1 用 α_1 說明

$$\therefore \frac{\partial t}{\partial t} = 1$$

加法節點的變化率

$$t = a + z \rightarrow \frac{\partial t}{\partial a} = 1$$

$$\hookrightarrow \frac{\partial t}{\partial b} = 1$$

$$\therefore z \text{ 的 grad} = 1$$

乘法節點 ...

$$a = x \cdot y$$

$$\frac{\partial a}{\partial x} = y, \quad \frac{\partial a}{\partial y} = x$$

$\hookrightarrow$ 再用鏈鎖率

$$\frac{\partial t}{\partial x} = \frac{\partial t}{\partial a} \cdot \frac{\partial a}{\partial x} = 1 \cdot y$$

$$\frac{\partial t}{\partial y} = \frac{\partial t}{\partial a} \cdot \frac{\partial a}{\partial y} = 1 \cdot x$$